IS 5960 MRP

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SAINT LOUIS UNIVERSITY

1. PROBLEM DEFINITION & SCOPE

SECTION	DESCRIPTION	YOUR RESPONSE
Problem Title	Give your problem a name. Focus on a system or process that is failing.	➢ Horizon State University – Student Retention Analysis
Problem description	Describe the problem in 3–4 sentences. Include what is going wrong, who is affected, and why it matters.	➢ The primary reasons for HSU's low first-year student retention rate include fragmented data and a lack of early risk identification tools.

1. PROBLEM DEFINITION & SCOPE

SECTION	DESCRIPTION	YOUR RESPONSE
Case-Bases Justification	Quote or paraphrase one part of the case that supports this problem. (Specific sentences found in the case you chose)	➢ Data dispersed among systems. ➢ There are no early or predictive alarms. ➢ Advisors take too late action. ➢ A greater number of dropouts.
Business Impact	What goals are affected (e.g., cost, efficiency, satisfaction)?	➢ loss of tuition income. ➢ A drop in reputation. ➢ The graduation rate is poorer. ➢ The advisor workload is inefficient.
Defined Scope	What system, team, process, or data stream will you focus on?	➢ Data integration. ➢ Constructing precise and comprehensible models. ➢ Dashboards that are easy to browse. ➢ Issues with security and privacy.
Expected Technology Involvement	List the IT/IS domains your solution will involve (e.g., data analytics, integration, automation).	➢ Provide advisers with early warnings and actionable insights by developing a predictive system with dashboards and connected data.

2. PROBLEM STATEMENT

In the first two years, Horizon State University's student retention rate is dropping due to disjointed data systems and a lack of early danger identification. Advisors' unclear perceptions of at-risk pupils result in protracted and frequently unsuccessful interventions. This results in financial losses, avoidable withdrawals, and worse academic success rates. Applying predictive analytics, integrating disparate data, and developing an easy-to-use platform that advisers can use are the challenges. An integrated predictive system that offers early warnings and insights will be designed as part of this project to enhance institutional stability and retention.

Defined Scope - HSU Early-Warning Student Retention System

- **User Group:** Academic advisors, retention specialists, and first-year program coordinators who monitor and support at-risk students.
- **Process Focus:** Integrating academic, financial, housing, and campus-engagement data to build a predictive model that flags students likely to drop out so advisors can intervene early.
- **Technology / System Type:** A centralized data platform with a predictive analytics engine and an advisor dashboard that provides risk scores, key drivers, and actionable alerts.
- **Boundaries / Limitations:**
  - Phase 1 limited to data integration from core campus systems and generation of early-warning risk scores with a basic advisor dashboard (MVP).
  - Focused on first-year undergraduate cohort to start, with the ability to scale to additional student groups and more complex interventions in later phases.
  - Does not initially include mobile apps, automated communication campaigns, or advanced AI explainability beyond top risk factors.

Business Alignment – HSU Early-Warning Student Retention System

- **Efficiency:** Cut advisor time spent manually gathering data from separate academic, financial, and housing systems by at least 40% through a unified predictive dashboard.
- **Student Experience:** Provide at-risk students with earlier, more targeted support so they receive timely academic and financial guidance, improving satisfaction and sense of belonging.
- **Retention & Revenue:** Reduce first-year dropouts and protect tuition income by enabling proactive interventions before students disengage.
- **Data Quality & Compliance:** Improve accuracy of student risk data and ensure that all analytics follow FERPA and institutional privacy policies.
- **Institutional Reputation:** Strengthen the university's image for student success and stability by demonstrating a data-driven approach to supporting first-year students.

Project Problem Statement

Horizon State University's first-year student retention rate is declining because the institution lacks a predictive early-warning system to identify students at risk of dropping out. Critical information—such as academic performance, financial status, housing records, and campus engagement—is stored in separate, unlinked systems, preventing advisors from gaining a complete and timely view of each student's situation. Without integrated risk indicators, interventions often occur only after a student has disengaged, leading to preventable tuition losses, reputational harm, and increased strain on advising staff. This project addresses the challenge by creating a centralized data platform that combines these disparate sources and generates early-warning risk scores. The initial scope focuses on core features—data integration, predictive modeling, and a simple advisor dashboard for timely alerts—to deliver immediate value and ensure compliance with privacy requirements such as FERPA. This Minimum Viable Product will provide a foundation for future phases that can expand to additional cohorts, introduce mobile access, and add advanced analytics to further enhance student persistence and institutional stability.

Project Problem Statement

Horizon State University's first-year student retention rate is declining because the institution lacks a predictive early-warning system to identify students at risk of dropping out. Critical information—such as academic performance, financial status, housing records, and campus engagement—is stored in separate, unlinked systems, preventing advisors from gaining a complete and timely view of each student's situation. Without integrated risk indicators, interventions often occur only after a student has disengaged, leading to preventable tuition losses, reputational harm, and increased strain on advising staff. This project addresses the challenge by creating a centralized data platform that combines these disparate sources and generates early-warning risk scores. The initial scope focuses on core features—data integration, predictive modeling, and a simple advisor dashboard for timely alerts—to deliver immediate value and ensure compliance with privacy requirements such as FERPA. This Minimum Viable Product will provide a foundation for future phases that can expand to additional cohorts, introduce mobile access, and add advanced analytics to further enhance student persistence and institutional stability.

MMP vs MVP

Phase	Features	Description	Ethical Considerations
MVP (Phase 1 – Minimum Viable Product)	Centralized Student Data Integration	Consolidate academic, financial, housing and engagement data into one secure repository to create a single, reliable view of each first-year student.	Comply with FERPA; encrypt all data; restrict access to authorized advising staff only.
MVP (Phase 1 – Minimum Viable Product)	Predictive Risk Scoring Dashboard	Apply machine-learning models to generate early-warning risk scores and show key drivers on an advisor dashboard.	Provide transparent model explanations; avoid bias in features; clearly communicate limits of prediction.
MMP (Phase 2 – Minimum Marketable Product)	Automated Intervention Workflows	Allow advisors to trigger standardized outreach (email, SMS) and track follow-up actions directly from the dashboard.	Obtain student consent for outreach; protect communication privacy; log advisor actions for accountability.
MMP (Phase 2 – Minimum Marketable Product)	Mobile and Expanded Cohort Access	Extend predictive analytics to other student groups and deliver risk dashboards on mobile devices.	Ensure secure authentication on mobile platforms and maintain the same privacy controls across all devices.

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The Technology Behind Our Solution

Data Management & Integration

We're combining scattered student data (from academics, finance, housing, etc.) into one unified repository, creating a single source of truth.

Business Intelligence & Visualization

An interactive **Microsoft Power BI dashboard** visualizes these risk scores, giving advisors clear, simple alerts they can act on immediately.

Predictive Analytics (AI/ML)

We use **Machine Learning** to analyze this data, identify at-risk patterns, and automatically generate predictive risk scores for each student.

Data Governance & Security

We ensure the entire system is safe, ethical, and trustworthy with strict **FERPA compliance** and role-based access to protect student privacy.

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